(iv) $x^3 - x = x(x^2 - 1)$

$= x(x^2 - 1^2)$

$= x(x+1)(x-1)$

(v) $x^4 - 16 = x^4 - 2^4 \quad \left[\because a^4 - b^4 = (a^2 + b^2)(a+b)(a-b)\right]$

$= (x^2 + 2^2)(x^2 - 2^2)$

$= (x^2 + 4)(x+2)(x-2)$

(vi) $x^2 + 4y^2 + 9z^2 - 4xy + 12yz - 6xz$

$= (-x)^2 + (2y)^2 + (3z)^2 + 2(-x)(2y) + 2(2y)(3z) + 2(3z)(-x)$

$= (-x + 2y + 3z)^2$ (or) $(x - 2y - 3z)^2$

Example 3.26

Factorise the following:

(i) $27x^3 + 125y^3$ (ii) $216m^3 - 343n^3$

(iii) $2x^4 - 16xy^3$ (iv) $8x^3 + 27y^3 + 64z^3 - 72xyz$

Solution

(i) $27x^3 + 125y^3 = (3x)^3 + (5y)^3 \quad \left[\because (a^3 + b^3) = (a+b)(a^2 - ab + b^2)\right]$

$= (3x + 5y)\left((3x)^2 - (3x)(5y) + (5y)^2\right)$

$= (3x + 5y)(9x^2 - 15xy + 25y^2)$

(ii) $216m^3 - 343n^3 = (6m)^3 - (7n)^3 \left[\because (a^3 - b^3) = (a-b)(a^2 + ab + b^2)\right]$

$= (6m - 7n)\left((6m)^2 + (6m)(7n) + (7n)^2\right)$

$= (6m - 7n)(36m^2 + 42mn + 49n^2)$

(iii) $2x^4 - 16xy^3 = 2x(x^3 - 8y^3)$

$= 2x\left(x^3 - (2y)^3\right) \quad \left[\because (a^3 - b^3) = (a-b)(a^2 + ab + b^2)\right]$

$= 2x\left((x - 2y)(x^2 + (x)(2y) + (2y)^2)\right)$

$= 2x(x - 2y)(x^2 + 2xy + 4y^2)$

(iv) $8x^3 + 27y^3 + 64z^3 - 72xyz$

$= (2x)^3 + (3y)^3 + (4z)^3 - 3(2x)(3y)(4z)$

$= (2x + 3y + 4z)(4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz)$

Thinking Corner

Check 15 divides the following

(i) $2017^3 + 2018^3$

(ii) $2018^3 - 1973^3$

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